DSGT:- Date :

Q4.)

→ Given:-

In a class 55 Student

Maths (M) = 23

Physics (P) = 24

chemistry (C) = 19

maths + Physics $(M \cap P) = 12$

maths + chemistry $(M \cap C) = 9$

Physics + chemistry $(P \cap C) = 7$

All three $(M \cap P \cap C) = 4$

(i) At least one Subject:-

→ Set formula

$$n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(M \cap C) - n(P \cap C) + n(M \cap P \cap C)$$

$$= 23 + 24 + 19 - 12 - 9 - 7 + 4$$

$$= 42$$

(ii) Exactly one Subject

$$n(\text{exactly one}) = n(M) + n(P) + n(C) - 2[n(M \cap P) + n(M \cap C) + n(P \cap C)] + 3[n(M \cap P \cap C)]$$

$$= 23 + 24 + 19 - 2(12 + 9 + 7) + 3(4)$$

$$= 23 + 24 + 19 - 2(28) + 12$$

$$= 66 - 56 + 12$$

$$= 22$$

(iii) Exactly two subject.

$n(\text{exactly two}) = n(M \cap P) + n(M \cap C) + n(P \cap C) - 3[n(M \cap P \cap C)]$

$= 12 + 9 + 7 - 3(4)$

$= 12 + 9 + 7 - 12$

$= 16$

Q5

Given :-

$T = 60\%$

$B = 50\%$

$J = 70\%$

$n(T \cap B) = 20\%$

$n(T \cap J) = 30\%$

$n(B \cap J) = 40\%$

let x = who play all three $[n(T \cap B \cap J)]$

Now,

use inclusion - ~~excl~~ exclusion principle.

$n(T \cup B \cup J) = n(T) + n(B) + n(J) - n(T \cap B) - n(T \cap J) - n(B \cap J) + n(T \cap B \cap J)$

$= 60 + 50 + 70 - 20 - 30 - 40 + x$

$n(T \cup B \cup J) = 90 + x$

So, The total percent can't ~~be~~ exceed 100%

$90 + x \leq 100$

$\boxed{x \leq 10}$

$\therefore$ The claim is not possible.

Q7.)

→ Given:-

Total television watcher = 500

Football (F) = 285

Hockey (H) = 195

Basketball (B) = 115

$n(F \cap B) = 45$

$n(F \cap H) = 70$

$n(H \cap B) = 50$

$n(\text{none}) = 50$

(i) How many pople watch all three games

Use inclusion-exclusion principle

$n(F \cup H \cup B) = n(F) + n(H) + n(B) - n(F \cap B) - n(F \cap H) + n(H \cap B) + n($

$500 = 285 + 195 + 115 - 45 - 70 - 50 + x$

$500 = 430 + x$

$500 - 430 = x$

$71 = x$

$x = 70$

But 50 do not watch all three games

$\therefore x = \cancel{70}\ \cancel{71 - 50}\ 70 - 50$

$x = 20$

$\boxed{x = 20}$: 20 people watch all three games.

(ii) How many people watch exactly one game?

→ $n(\text{exactly one}) = n(F) + n(H) + n(B) - 2[n(F \cap B) + n(F \cap H) + n(H \cap B)] + 3[(F \cap H \cap B)]$

$= 285 + 195 + 115 - 2(45 + 70 + 50) + 3(20)$

$= 285 + 195 + 115 - 2(165) + 60$

$= 28 + 195 + 115 - 330 + 60$

$n(\text{exactly one}) = 325$

UNIT – 3

Q1

→

Given:-

$A = \{a, b, c, d\}$

$B = \{1, 2, 3\}$.

Determine whether the relation R from A to B is a function

(i) $R = [(a,1)\ (b,2)\ (c,1),\ (d,2)]$

→ $a \to 1$ $\quad c \to 1$

$b \to 2$ $\quad d \to 2$

every element of A has exactly one image in B.

So, R is a function.

Range $= \{1, 2\}$.

(ii) $R = [(a,1)\ (b,2)\ (a,2)\ (c,1)\ (d,2)$

→ $a \to 1$ & $a \to 2$

$b \to 2$ $\quad c \to 1$

$d \to 2$

Since A has two different value

So R is not a function.

Q2

→ Given:-

$R = \{(1,4)\ (2,3)\ (3,10)\ (4,8)\ (5,6)$

Check whether $R_1\ R_2\ R_3\ R_4$ are function or not.

① $R_1 = \{(1,4)\ (2,4)\ (3,4)\ (4,4)\ (5,4)\}$

→ each element has exactly one elements.
so, it is a function

② $R_2 = \{(1,2), (2,4)\ (2,10)\ (3,8)\ (4,6)\ (5,4)\}$.

→ The elemat are ~~not~~ not in different 2 is repeated..
so, it is not function.

③ $R_3 = \{(1,6)\ (2,2)\ (4,4)\ (5,10)\}$

Each element has exactly one element
So, it is a function.

④ $R_4 = \{(1,6)\ (2,2)\ (3,2)\ (4,4)\ (5,10)\}$

Each elements are exactly have one element
So, it is a function

Q 3.7

function $f(n) = \frac{(4x+3)}{(5x-2)}$ find f^{-1}

$$f(x) = \frac{(4x+3)}{(5x-2)}$$

Replace $f(x)$ with y

$$y = \frac{4x+3}{5x-2}$$

Swap x and y

$$x = \frac{4y+3}{5y-2}$$

solve to y.

$$x(5y-2) = 4y+3$$

$$5xy - 2x = 4y+3$$

$$5xy - 4y = 3 + 2x$$

$$y(5x-4) = (2x+3)$$

$$y = \frac{2x+3}{5x-4}$$

$$\boxed{f^{-1} = \frac{2x+3}{5x-4}}$$

Q4
→

(a) Each person in the earth is assigned a number which corresponds to ~~this~~ His age

→ function - Yes
Injective - No.
Surjective - No
Bijective - No.

(b) Each Student is assigned a teacher

→ function - Yes each student assign teacher (assuming)
Injective - No
Surjective - No
Bijective - No.

(c) Each country has assigned it's capital.

→ function - Yes
Injective Yes
Surjective - No
Bijective - No.

Q5

→

① check if f is one to one (Injective)

→ let $f(a) = f(b)$

$a^3 + a + 1 = b^3 + b + 1$

$a^3 - b^3 + a - b = 0$

$(a-b)(a^2 + ab + b^2 + 1) = 0$

$a - b = 0$

$\therefore a = b$.

Hence f is injective (one to one)

② che if f is onto (surjective)

$y = x^3 + x + 1$

x	$f(x$
-3	-29
-2	-9
-1	-1
0	1
1	3
2	11
3	31

Not every integer is hit (e.g 2, 4, 6, 8)...

∴ so f is not Surjective (onto)

③ $f(x) = x^3 + x - 1$ is one-to-one but not onto (hence not bijective).

Define

① Planner Graph.

A planner Graph can be represented is Such a way that no two edges intersect or overlap except where they meet at a common vertex.

Example. A triangle or square planner Graph.

② Cut vertices

A cut vertex is a vertex that, if deleted. breaks the graph into two or more parts.

Example A - B - C

~~B is~~ B is cut vertex becaues if we remove B the graph breaks into two Separate vertice A and C

③ chain

A chain is a path that links one vertex to another through a Sequence of connected edge.

④ monoid.

~~A chain is a list~~ A monoid is like a Semigroup (a set with associative operation), but it also has Identity element.

5. Group:-

A Group is a collection of elements with a way to combine them that follows 4 rule

① closure
② Associative
③ Identity
④ Inverse

UNIT - IV

Q1.

If you have n bicycle and k color at least one box will contain at least $\left\lceil \frac{n}{k} \right\rceil$ item.

Total bycycle = 50 → n = 50

Total color = 7 → k = 7

If we have 7 bicycle per color and 7 color

total 7×7 = 49.

But we have 50 bicycle which is 1 more than 49.

So,

This is a direct application of a pigeonhole Principle

Q2

→ Apply pigeonhole principle.

If we distribute friends into 12 months the worst-case ~~as~~ scenario is that birthday are spread out evenly as possible.

~~Th~~

Each mont could have up to 4 fren without any month reaching 5

so, if each of the 12 month had 4 friends

$12 \times 4 = 48$

As soon as we add one more friend

$48 + 1 = 49.$

you need 49 friends to gurantee that at least 5 have birthday in the same month

UNIT-V

Q1

→ $(Z, *)$ is an abelian group

① closure

$a*b = a+b+1$

$a*b = z$

② Associative

we must show

$(a*b)*c = a*(b*c)$

compute LHS

$(a*b)*c = (a+b+1)*c = (a+b+1)+c+1 = a+b+c+2$

compute RHS

$a*(b*c) = [a*(b+c+1)] = a+(b+c+1)+1 = a+b+c+2.$

$(a*b)*c = a*(b*c)$

③ Identity

we must show

$a*e = a$

compute

$a*e = a+e+1 = a.$

$e+1 = 0 \quad e = -1$

④ Inverse

we must show

$a*a^{-1} = e$

compute

$a*a^{-1} = a+a^{-1}+1 = -1$

⑤ commutative

$a*b = b*a$

$a*b = a+b+1 = b+a+1 = b*a.$

$a*b = b*a$